Yehudah Ackermann

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Profile

M.Sc. student in aerospace engineering at the Technion, focusing on air-breathing rotating detonation combustors. Before that, two years building precision cryogenic and vacuum hardware for quantum-technology research, from design and machining to integration and test. Quick to generate workable design concepts and carry them through to hardware; strongest where physics, hardware and data processing meet.

Research

Technion, Faculty of Aerospace Engineering **Haifa**
M.Sc. Thesis: Rotating Detonation Scramjet *2026–present*

- Integrating a rotating detonation combustor into a scramjet flow path, end to end: theory, design, simulation, fabrication and testing.
- CFD of supersonic mixing and detonation (OpenFOAM), validated against published experiments.
- Instrumentation and data acquisition for high-speed combustion diagnostics.

Experience

Simon Fraser University, Silicon Quantum Technology Group **Burnaby, Canada**
Research Technician *Jul 2022–Jul 2024*

- Designed, machined and integrated custom hardware for cryogenic and vacuum systems (1 K, wet/dry cryostats).
- Built precision calibration, magnetic-field and microwave hardware to tight specifications.
- Set up and maintained lab infrastructure: high vacuum, helium handling and mechanical systems.

Additional Experience & Projects

2022–24: **Photonic Inc., Research Assistant**: rapid-prototyping tools for liquid-helium setups; automated verification of DC, RF and microwave signal lines at cryogenic temperatures.

2021: **Canadian Nuclear Laboratories, Applied Physics Intern**: algorithms to determine uranium enrichment from gamma-ray spectra.

2020: **McMaster University, Undergraduate Researcher**: spectral discretization methods in MATLAB (Chebfun); co-authored an open-access paper on high-precision function computation.

2021–22: **Automatic bike-wheel truing stand**: led a 4-person team combining FE modelling, machine vision and mechatronic control for sub-millimetre alignment; won Best Use of Mechanical Design.

2020: **Bluetooth indoor positioning**: multi-microcontroller (MQTT) system for room-level location from RSSI mapping.

Education

Technion – Israel Institute of Technology **Haifa**
M.Sc. Aerospace Engineering *2026–2028*

McMaster University **Hamilton, Canada**
B.Eng. Engineering Physics, Minor in Applied Mathematics *2017–2022*
Dean's Honour List (2020) | Stewart Fund Award

Skills

Programming: Python, MATLAB, C, C++, Maple, APDL; Linux, Git, LaTeX

Simulation & CAD: OpenFOAM, ANSYS (Mechanical, CFX, Fluent), CalculiX, Gmsh, SolidWorks, Fusion 360

Hands-on: Manual machining, 3D printing (FDM/SLA), vacuum systems, cryogenics, electronics

Languages: English (fluent), Hebrew (basic). Made Aliyah in September 2025. References on request.